



UNIVERSITI TEKNOLOGI MARA (UiTM), PUNCAK PERDANA CAMPUS

FACULTY OF INFORMATION SCIENCE

**BACHELOR OF INFORMATION SCIENCE (HONS) INFORMATION SYSTEM MANAGEMENT
(CDIM262)**

ADVANCED WEB DESIGN DEVELOPMENT AND CONTENT MANAGEMENT (IMS566)

GROUP ASSIGNMENT:

Web Application Project – MINDEASE

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TABLE OF CONTENTS

1.0	INTRODUCTION	1
2.0	GITHUB REPOSITORY LINK	2
3.0	ENTITY-RELATIONSHIP DIAGRAM (ERD)	2
4.0	SYSTEM REQUIREMENTS	5
4.1	SOFTWARE	5
4.2	VERSION COMPATIBILITY	5
4.3	DEPENDENCIES	6
5.0	USER INTERFACE OVERVIEW	7
5.1	OVERALL LAYOUT STRUCTURE	7
5.2	COLOUR SCHEME AND BRANDING	7
5.3	NAVIGATION SYSTEM	8
5.4	STATUS BADGES AND RISK COLOR CODING	10
6.0	FEATURES AND FUNCTIONALITIES	11
6.1	USER AUTHENTICATION MODULE	11
6.1.1	Login Page	11
6.1.2	Registration	12
6.1.3	User Profile Management	13
6.2	WELLNESS ASSESSMENT MODULE	15
6.2.1	Assessment Form (For Student)	15
6.2.2	Auto-Scoring & Risk Engine	18
6.2.3	Assessment Result Page & Crisis Alert	19
6.2.4	My History (For Student)	21
6.2.5	All Assessments (For Counselor & Admin)	23
6.3	COUNSELOR NOTES MODULE	25
6.4	QUESTIONS MANAGEMENT MODULE (ADMIN ONLY)	28
6.5	DASHBOARD MODULE (ROLE-AWARE)	30
6.6	PDF REPORT GENERATION	32
6.7	USERS MANAGEMENT AND SITE CONFIGURATION (ADMIN ONLY)	34
6.8	INFORMATION PAGES	35
6.8.1	FAQ Page	35
6.8.2	Contact Us Page	36
7.0	WORKFLOW OF FORM (ILLUSTRATION OF THE CRUD CYCLE)	38

7.1	CRUD WORKFLOW DIAGRAM.....	38
7.2	DETAILED CRUD PROCESS	38
7.3	CRUD OPERATIONS FOR OTHER MODULES	39
8.0	TEAM ROLES AND CONTRIBUTIONS.....	41
9.0	CONTACT INFORMATION	42
9.1	SYSTEM CONTACT.....	42
9.2	CRISIS SUPPORT RESOURCES	42
9.3	ACADEMIC CONTACT	42
10.0	CONCLUSION AND REFLECTION	43
10.1	PROJECT SUMMARY	43
10.2	TECHNICAL ACHIEVEMENTS.....	43
10.3	CHALLENGES FACED AND SOLUTIONS.....	44
10.4	FUTURE IMPROVEMENTS	45
	REFERENCES	47

1.0 INTRODUCTION

MindEase is web-based mental wellness develop for Universiti Teknologi MARA (UiTM) students. This system now digitizes from traditional paper-based counselling intake forms and student wellness assessments has allowing students to complete structured online self-assessments of their mental wellbeing while enabling university counselors to monitor, evaluate and follow up on student wellness systematically.

The main purpose of this system to provide a personal and easily accessible channel for students to check their mental wellbeing. Student complete a guided clinical assessment with 25 questions based on three internationally recognized screening instruments they are PHQ-9 (Patient Health Questionnaire for depression), GAD-7 (Generalized Anxiety Disorder scale) and PSS-4 (Perceived Stress Scale) along with a customized General Wellbeing section covering sleep, academic stress, social support, loneliness and physical activity.

What sets MindEase apart from a simple form collection system is its automatic scoring and risk engine. Once submitted, the system instantly calculates the PHQ-9, GAD-7 and PSS-4 scores, classifies the severity level for each instrument and determines the overall risk level. **Any indication of self-harm will immediately trigger a crisis alert** displaying emergency hotline numbers which are Talian Kasih 15999, Befrienders KL and UiTM Counseling Unit also flag the case at the top of the counselor's dashboard for immediate review.

The system implements role-based access control with three user roles. Student can submit assessments, view their own results and history and read counselor feedback. Counselors can view all submissions, filter and search by risk level, add clinical notes with follow-ups and update case status. Meanwhile the administrator has full access, including user management, question banks, system configuration and PDF report generation.

MindEase is build on the reCRUD framework, a fast CRUD development stacks (Apache, MySQL, PHP). The interface uses a soothing purple color scheme inline with the mental wellness theme and it is fully responsive so students can check in with the counsellors. The system also comes with support information pages including an FAQ page with mental wellness questions and a Contact Us page where students can reach out to the counseling unit.

2.0 GITHUB REPOSITORY LINK

Link: <https://github.com/meanryncoding/mindease.git>

3.0 ENTITY-RELATIONSHIP DIAGRAM (ERD)

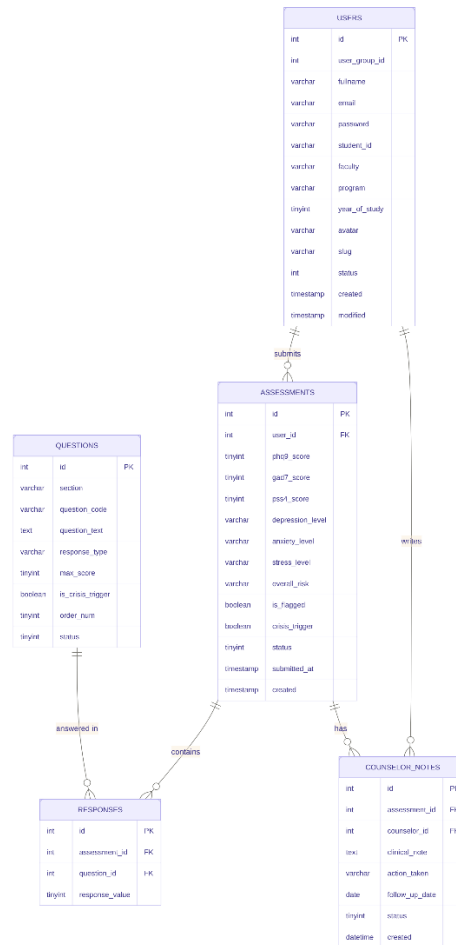


Figure 3.0: ERD of MindEase

The MindEase database includes of five related tables:

Table	Purpose	Key Relationship
User	Stores all user accounts. The role is determined by the user_group_id field (admin,	One user (student) submits many assessments; one user

	counselor, student). We extended the default reCRUD table with academic profile fields: student_id, faculty, program, and year_of_study. Other reCRUD built-in fields include avatar, slug, and status.	(counselor) writes many counselor_notes.
Questions	The question bank containing all 25 assessment questions (PHQ-9, GAD-7, PSS-4, and Section D), each with a question_code, section, question_text, response_type, max_score, order_num, and status. The is_crisis_trigger flag identifies which question (A9) activates the crisis alert when answered above 0.	One question is answered in many responses.
Assessments	One record per submitted assessment; stores the computed phq9_score, gad7_score, pss4_score, severity levels (depression_level, anxiety_level, stress_level), overall_risk, is_flagged, crisis_trigger, status, and submitted_at timestamp.	Belongs to one user; contains many responses; has many counselor_notes.
Responses	Stores each individual answer given by the student, linked to both the assessment and the question via a single tinyint	Belongs to one assessment and one question.

	response_value column. All sections including Section D use numeric codes as the stored value.	
Counselor_notes	Clinical notes added by counselors on a specific assessment, including clinical_note, action_taken, follow_up_date, and status. The counselor_id links the note back to the counselor who wrote it.	Belongs to one assessment and one counselor (user).

Each table has own independent auto-increment primary key (PK), and records are linked through foreign keys(FK) such as users.id → assessments.user_id (submits), users.id → counselor_notes.counselor_id (writes), assessments.id → responses.assessment_id (contains), assessments.id → counselor_notes.assessment_id (has), questions.id → responses.question_id (answered in). In addition to the five core tables, the system uses reCRUD's built-in supporting tables, including faqs for the FAQ and contacts for Contact Us submissions.

4.0 SYSTEM REQUIREMENTS

4.1 SOFTWARE

This software required to run the MindEase system:

Category	Software	Description
Operating System	Windows 11	The development and deployment environment
Web Server Stack	Laragon 2026	Includes Apache web server, MySQL, and PHP
Database Server	MySQL 8.4	For storing all system data
PHP Runtime	PHP 8.3	Server-side scripting language
Framework	reCRUD (CakePHP 4 based)	Rapid CRUD development boilerplate used as the project base
Package Manager	Composer 2.x	PHP dependency manager
Database Tool	phpMyAdmin	For managing tables, columns, and seed data
Web Browser	Google Chrome	Latest versions recommended



Figure 4.1: Laragon running with Apache and MySQL started

4.2 VERSION COMPATIBILITY

The MindEase system has been tested and verified to work with the software versions

Component	Minimum Version	Recommended Version
PHP	7.4	8.1 or higher

CakePHP Framework (via reCRUD)	4.x	4.x (latest)
MySQL	5.7	8.0
MariaDB	10.3	10.6
Composer	2.0	2.6 or higher
Apache	2.4	2.4

4.3 DEPENDENCIES

The MindEase system relies third-party libraries and package

Core Dependencies

Package	Version	Purpose
cakephp/cakephp	^4.x	Main PHP framework providing MVC architecture (bundled in reCRUD)
cakephp/authentication	^2.x	Handles user login, logout, and session management
friendsofcake/cakepdf (DomPDF engine)	^4.x	PDF generation for wellness reports
lorenzo/audit-stash	^4.x	Audit logging of user actions (built into reCRUD)
cakephp/migrations	^3.x	Database migration tool

Libraries

Library	Purpose
Bootstrap 5	Responsive layout, cards, tables, and form styling (bundled with reCRUD templates)
Chart.js	Dashboard data visualisations (monthly assessment trend chart)
Font Awesome	Icon library for the sidebar, buttons, and status badges
Vanta.js	Animated 3D cloud background on the login page, enhancing the calming visual theme of the system

5.0 USER INTERFACE OVERVIEW

The MindEase system features give a clean, calm, interactive and responsive user interface designed to make a sensitive task but not too sensitive due to reporting on one's mental wellbeing so user can feel private, safe and non-intimidating.

5.1 OVERALL LAYOUT STRUCTURE

The system uses a consistent the dashboard style throughout all pages with following components:

1. **Sidebar** - Including MindEase logo featuring a colour-cycling animated icon the navigation menu split into two groups, a top section which are Dashboard, FAQ, Contact Us and Profile. It visible to all roles, plus Documents for admin only and a role section below STUDENT / COUSELOR / ADMINISTRATOR with role-specific menu items.
2. **Top Bar** - Displaying hamburger toggle to collapse the sidebar, the page title and action icons on the right including dark/light mode toggle, fullscreen and the logout button.
3. **Main Content Area** - Displaying current page content in Bootstrap cards with proper spacing

5.2 COLOUR SCHEME AND BRANDING

The system uses a green and white colour scheme to the main interface. The sidebar navigation and active menu highlights are rendered in green, while the MindEase logo features a continuously color-cycling animated icon. Purple accents are used strategically in high-visibility areas when most prominently the student dashboard welcome banner. This colour scheme is applied consistently across all pages.

Element	Color	Code / Style
Logo Icon	Animated color-cycling	CSS@keyframes hue-rotate animation – cycles through multiple hues continuously.
Active Menu Highlight	Light Green	Bootstrap success tint
Student Banner Accent	Deep Purple	#2d1b69 to #5b2d8e gradient

Background	Light Neutral	Bootstrap body-tertiary (white/light grey)
Risk - Low / Minimal	Green	Badge (success)
Risk - Mild	Yellow	Badge (warning)
Risk - Moderate	Orange	Badge (orange)
Risk - High / Critical	Red	Badge (danger)

5.3 NAVIGATION SYSTEM

The sidebar navigation adapts based on the logged-in user role (by user_group_id):

The sidebar is divided into two parts. The top section have items common to all roles (include Documents visible to admin only). Each role has its own labelled section with role-specific menu items.

Student Sidebar	Counselor Sidebar	Admin Sidebar
TOP SECTION (all roles)	TOP SECTION (all roles)	TOP SECTION (admin)
Dashboard	Dashboard	Dashboard
FAQ	FAQ	FAQ
Contact Us	Contact Us	Contact Us
Profile	Profile	Documents
		Profile
STUDENT section	COUNSELOR section	ADMINISTRATOR section
New Assessment	All Assessments	All Assessments
My History	Counselor Notes	Counselor Notes
		Questions (Question Bank)
		Site Configuration
		User Management

The logout button is located in the top bar (*penjuru kanan atas*), not in the sidebar. The Documents module is visible to admins only and hidden from students and counselors as it is not relevant to the student wellness workflow. Administrative modules such as Questions, Site Configuration and User Management are restricted to admin only. Student who attempt to access a restricted URL directly are redirected to the assessment form.

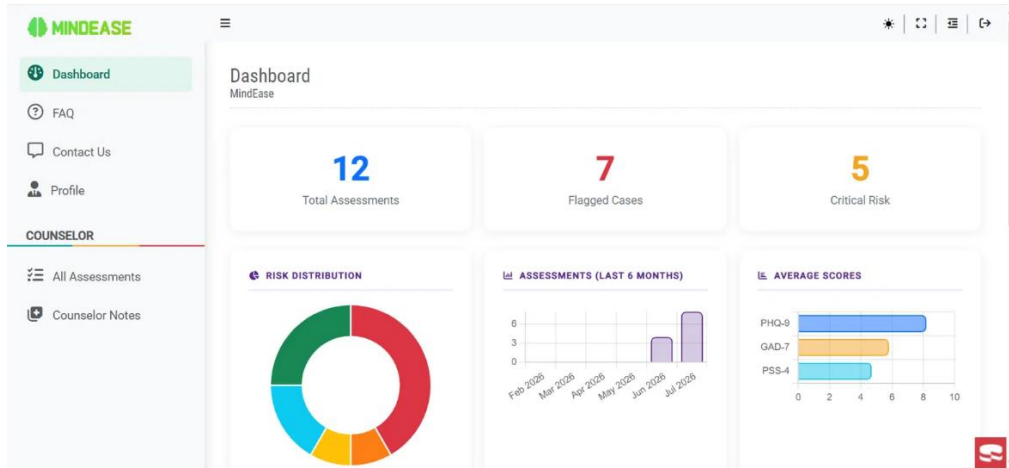


Figure 5.3.1: sidebar navigation menu for counselor

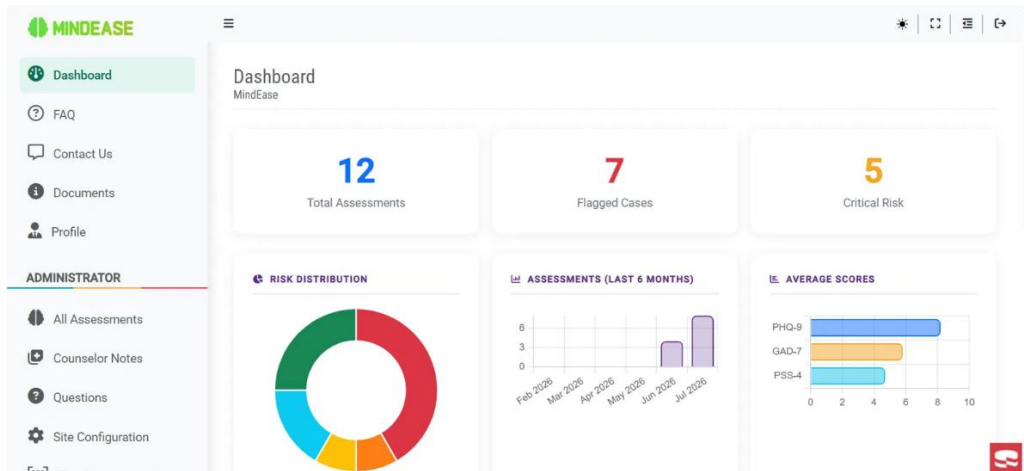


Figure 5.3.2: sidebar navigation menu for admin

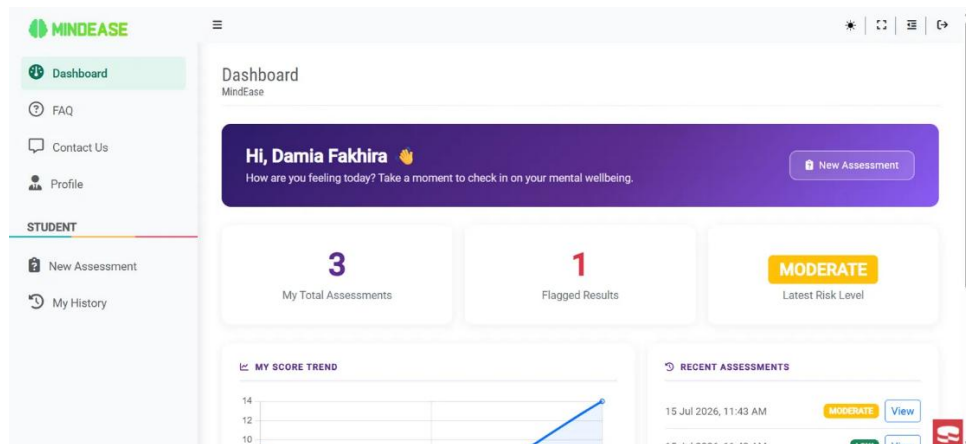


Figure 5.3.3: sidebar navigation menu for student

5.4 STATUS BADGES AND RISK COLOR CODING

Throughout the system, risk levels and case statuses are shown as colour coded badges and icons so that a counselor can triage at a glance

- **Risk Level Badges** - CRITICAL (red), HIGH (dark red), MODERATE (yellow), MILD (cyan), LOW (green)
- **Special Icons** – Flagged cases display a warning icon, Crisis cases display a alert icon
- **Review Status** – Reviewed (green checkbox), Pending Review (hourglass icon)

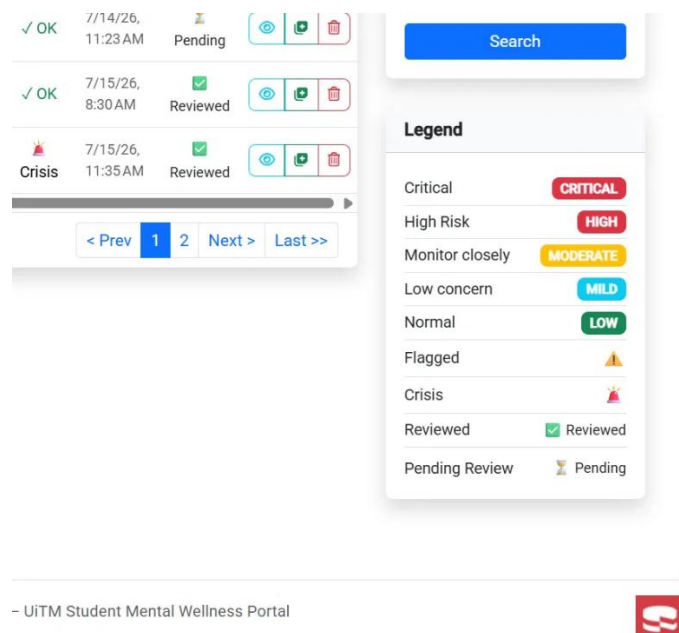


Figure 5.4: Legend panel showing all colour coded risk badges and status icons in the assessments list

6.0 FEATURES AND FUNCTIONALITIES

6.1 USER AUTHENTICATION MODULE

6.1.1 Login Page

The login page was fully redesigned from the default reCRUD login into a modern split-panel layout. The left panel carries the MindEase branding on a purple gradient background with the system tagline and a short feature list while the right panel contains the login form. An animated cloud background (Vanta.js) fills the area outside the card, reinforcing the calm and welcoming atmosphere.

Key features:

- Split-panel design: purple gradient branding panel (left) and login form (right)
- Email and password login with secure password hashing
- Session management handled by the CakePHP Authentication plugin
- Role detection using the identity's `user_group_id` to control menu visibility
- "Create Account" link for new student self-registration
- "Forgot Password?" link for account recovery
- Quick links below the form: FAQ and Contact Us (accessible even before login)
- Flash error message for invalid credentials

How to use:

1. Open the MindEase login page
2. Enter your registered email and password
3. Click the **Sign In** button
4. You are redirected to the dashboard for your role

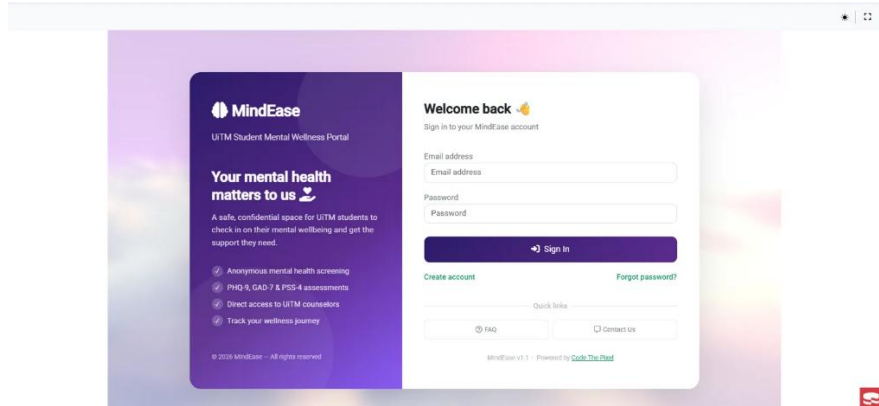


Figure 6.1.1: Redesigned split-panel login page with Vanta.js cloud background and purple branding panel.

6.1.2 Registration

New students can create an account through the User Registration page. Accounts are automatically assigned the Student role, while Counselor and Admin accounts are created by the administrator through Users Management.

Registration fields:

- Fullname
- Email
- Password (with confirmation)
- Profile Image (optional file upload for avatar)
- "I agree to the registration term & conditions" checkbox

Key notes (displayed on the right panel):

- All information provided during registration is correct
- One email can register only one account
- Please use a strong password to ensure your account is secure
- Contact administrator if unable to register

How to use:

1. Click the **Create Account** link on the login page
2. Fill in your full name, email address, password, and confirm password

3. Upload a profile image if desired
4. Check the "I agree to the registration term & conditions" checkbox
5. Click **Submit** to create your account
6. Log in with your new credentials, then complete your academic details (Student ID, Faculty, Program, Year of Study) in the Profile page

Figure 6.1.2: User Registration page showing registration form and notes panel

6.1.3 User Profile Management

Logged-in users can view and update their profile through the Account Details page. The page was redesigned with a purple gradient banner header showing the user's profile photo, name, role badge, and active status. The layout adapts based on the logged-in user's role.

Profile Page Features (all roles):

- Four tabs: **Account**, **Update**, **Password**, and **Activities**
- Purple gradient banner with profile photo, name, role, and Active status badge
- **Account Information** panel — Full Name, Email, Group, Status, Verified badge, Created on date
- **Profile QR** panel — scannable QR code linking to the user's profile page

Student profile only — additional panel:

- **Student Information** panel — Student ID, Faculty, Program, and Year of Study (conditionally rendered, hidden for Counselors and Admins)

Editable Fields (Update tab):

- Full Name and Email Address

- Student ID, Faculty, Program, Year of Study (*student only*)

Implementing this feature required three coordinated changes: (1) adding the columns to the users table with an ALTER TABLE statement (student_id, faculty, program, year_of_study); (2) adding the new form controls to the profile Update template; and (3) whitelisting the four new fields in the User entity's _accessible array so CakePHP's mass-assignment protection would allow them to be saved. The post-save redirect was also changed from the referer to the profile page so users are not redirected to a restricted page.

How to Use:

1. Log in and click **Profile** in the sidebar
2. Click the **Update** tab
3. Fill in or edit your details — students can also update Student ID, Faculty, Program, and Year of Study
4. Click **Submit** — you are returned to the Account tab with updated details shown
5. To change your password, use the **Password** tab instead
6. The **Activities** tab shows your recent system activity log

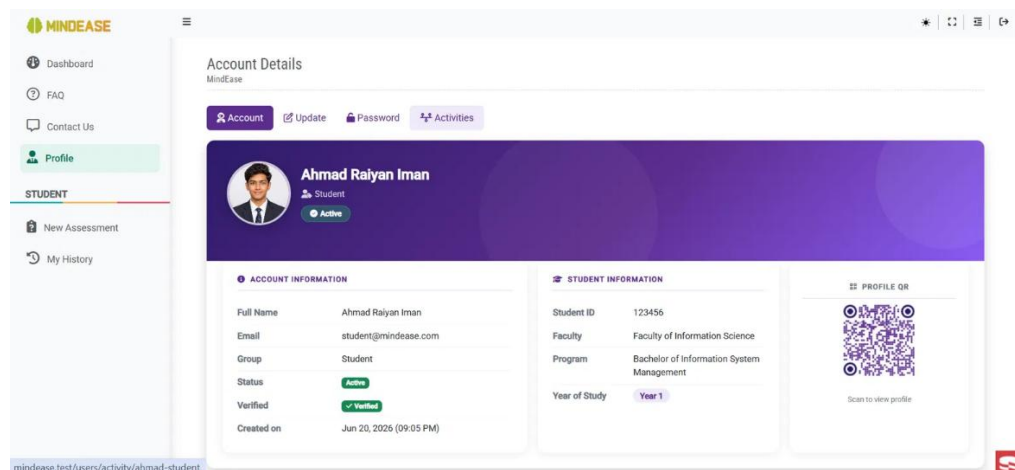


Figure 6.1.3.1: Student profile page

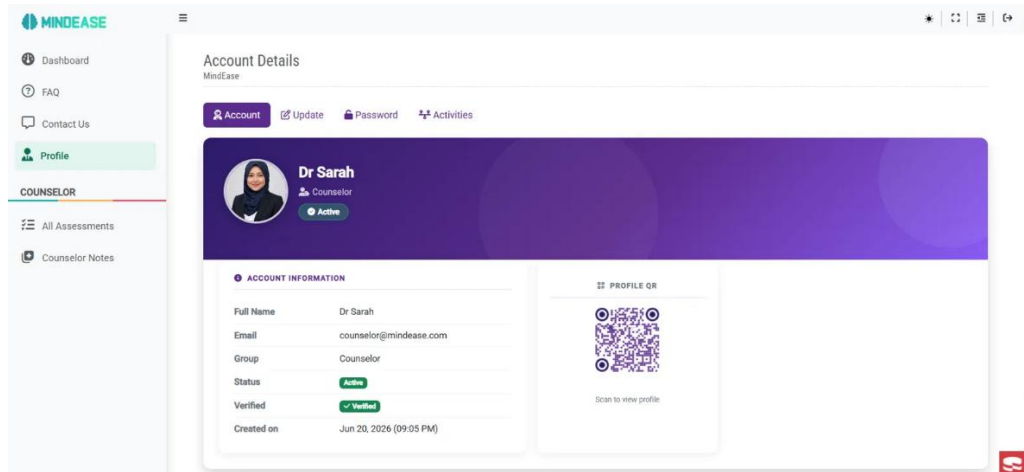


Figure 6.1.3.2: Counselor profile page

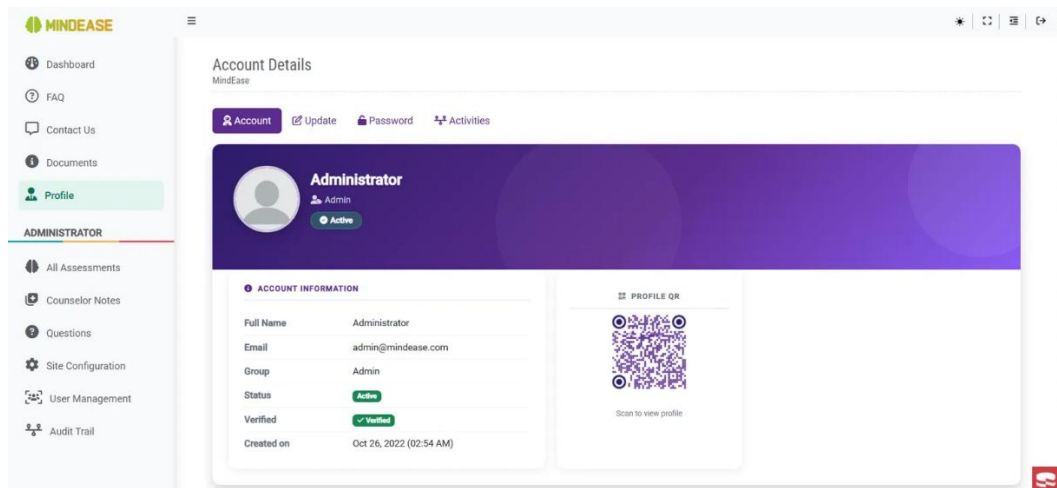


Figure 6.1.3.3: Admin profile page

6.2 WELLNESS ASSESSMENT MODULE

6.2.1 Assessment Form (For Student)

The core of MindEase is the 25-question clinical assessment form, accessible via the **"How Are You Feeling Today?"** page. The form is organised into four sections with a **4-step progress indicator** at the top showing A → B → C → D. Completed steps turn dark green, the active step is highlighted in green, and upcoming steps remain grey.

Each section has its own **colour-coded banner header** to visually distinguish the sections

Section	Instrument	Banner Colour	Response Type
A — Depression	PHQ-9 (9 questions)	Blue	4 button cards: Not at all (0) / Several days (1) / More than half (2) / Nearly every day (3). Question A9 screens for self-harm ideation.
B – Anxiety	GAD-7 (7 questions)	Orange	4 button cards: same options as Section A. Selected answer highlights in green.
C – Stress	PSS-4 (4 questions)	Cyan	4 button cards: Never (0) / Almost never (1) / Sometimes (2) / Fairly often (3). Selected answer highlights in dark green.
D – General Wellbeing	Custom (5 questions)	Green	Dropdown select menus (e.g. sleep hours: Less than 4 hours / 4–6 hours / 6–8 hours / More than 8 hours). Does not affect clinical scores.

All questions are stored in the questions table (question bank) rather than hardcoded in the form. The form template loops through the question bank grouped by section, which means questions can be edited or extended by the admin without touching the code, and every response is stored in the responses table linked to the assessment by assessment_id and to the question by question_id, as a tinyint response_value.

How to Use:

1. Log in as a student and click **New Assessment** in the sidebar
2. Answer all questions in Section A (PHQ-9) by clicking one button card per question
3. Continue through Section B (GAD-7), Section C (PSS-4) — the progress indicator at the top fills green as you advance
4. For Section D (General Wellbeing), select an answer from each dropdown menu
5. Click **Submit Assessment**
6. You are redirected to your result page showing your scores and overall risk level; if a crisis is detected, emergency hotline numbers are displayed immediately

Section A — Depression Screening (PHQ-9)
Over the last 2 weeks, how often have you been bothered by any of the following?

A1 Little interest or pleasure in doing things

Not at all (0) Several days (1) More than half (2) Nearly every day (3)

A2 Feeling down, depressed, or hopeless

Not at all (0) Several days (1) More than half (2) Nearly every day (3)

A3 Trouble falling or staying asleep, or sleeping too much

Not at all (0) Several days (1) More than half (2) Nearly every day (3)

Figure 6.2.1.1: Section A — Depression Screening (PHQ-9)

Section B — Anxiety Screening (GAD-7)
Over the last 2 weeks, how often have you been bothered by the following problems?

B1 Feeling nervous, anxious, or on edge

Not at all (0) Several days (1) More than half (2) Nearly every day (3)

B2 Not being able to stop or control worrying

Not at all (0) Several days (1) More than half (2) Nearly every day (3)

B3 Worrying too much about different things

Not at all (0) Several days (1) More than half (2) Nearly every day (3)

Figure 6.2.1.2: Section B — Anxiety Screening (GAD-7)

Section C — Stress Screening (PSS-4)
In the last month, how often have you...

C1 Been upset because of something that happened unexpectedly

Never (0) Almost never (1) Sometimes (2) Fairly often (3)

C2 Felt unable to control the important things in your life

Never (0) Almost never (1) Sometimes (2) Fairly often (3)

C3 Felt nervous and stressed

Never (0) Almost never (1) Sometimes (2) Fairly often (3)

Figure 6.2.1.3: Section C — Stress Screening (PSS-4)

Figure 6.2.1.4: Section D — General Wellbeing

6.2.2 Auto-Scoring & Risk Engine

When the student submits the form, the AssessmentsController automatically computes all scores and determines the risk level before saving. No manual computation by the counselor is required.

Score Computation:

- PHQ-9 Score = A1 + A2 + ... + A9 (maximum 27)
- GAD-7 Score = B1 + B2 + ... + B7 (maximum 21)
- PSS-4 Score = C1 + C2 + C3 + C4 (maximum 16)

PHQ-9 Scoring Matrix (Depression):

Score Range	Depression Level	System Action
0 - 4	Minimal	No flag
5 - 9	Mild	No flag
10 - 14	Moderate	Flag to counselor
15 - 19	Moderately Severe	High risk alert
20 - 27	Severe	Critical alert

GAD-7 Scoring Matrix (Anxiety):

Score Range	Anxiety Level	System Action
0 - 4	Minimal	No flag
5 - 9	Mild	No flag
10 - 14	Moderate	Flag to counselor

15 - 21	Severe	Critical alert
---------	--------	----------------

PSS-4 Scoring Matrix (Stress):

Score Range	Stress Level	System Action
0 - 4	Low	No flag
5 - 8	Moderate	No flag
9 - 12	High	Flag to counselor
13 - 16	Very High	High risk alert

Overall Risk Determination Logic:

1. IF A9 > 0 (any self-harm ideation) the assessment is immediately classified as CRITICAL and the crisis trigger is set, regardless of total scores
2. ELSE IF PHQ-9 >= 15 OR GAD-7 >= 15, the overall risk is HIGH and the case is auto-flagged
3. ELSE IF PHQ-9 >= 10 OR GAD-7 >= 10, the overall risk is MODERATE and the counselor is notified
4. ELSE IF PHQ-9 >= 5 OR GAD-7 >= 5, the overall risk is MILD (monitored)
5. ELSE the overall risk is LOW

The computed scores, severity levels, overall risk, is_flagged flag, and crisis_trigger flag are stored directly in the assessments table. Storing scores at submission time avoids recomputing them from the responses table every time the counselor dashboard loads, keeping list queries fast.

6.2.3 Assessment Result Page & Crisis Alert

After submission, the student is redirected to the **Assessment Details** page, which presents the outcome in a clear clinical layout. The page content adapts depending on whether a crisis was detected.

If crisis is detected (question A9 answered above 0):

A **Crisis Alert Detected!** banner appears at the very top with a pink background containing:

- Message: *"This student has indicated thoughts of self-harm. Please seek help immediately."*

- Three emergency contacts in bold:
 - **Talian Kasih:** 15999
 - **Befrienders KL:** 03-76272929
 - **UiTM Counseling:** Contact your faculty counselor

All submissions (always displayed):

- **Overall Risk Level** card — shows risk level (e.g. CRITICAL, HIGH, MODERATE, MILD, LOW) with submission timestamp
- **Flagged — Counselor Review Required** badge — shown if the case has been escalated
- **Three score cards** displayed side by side:
 - PHQ-9 — Depression (blue, score out of 27, severity badge)
 - GAD-7 — Anxiety (orange, score out of 21, severity badge)
 - PSS-4 — Stress (teal, score out of 16, severity badge)
- **Full response table** — shows the student's answers to all 25 questions
- **Counselor Notes section** — visible once a counselor has added a note on this submission

The same submission appears at the top of the counselor's All Assessments list with a CRITICAL badge and icon so it can be actioned urgently.

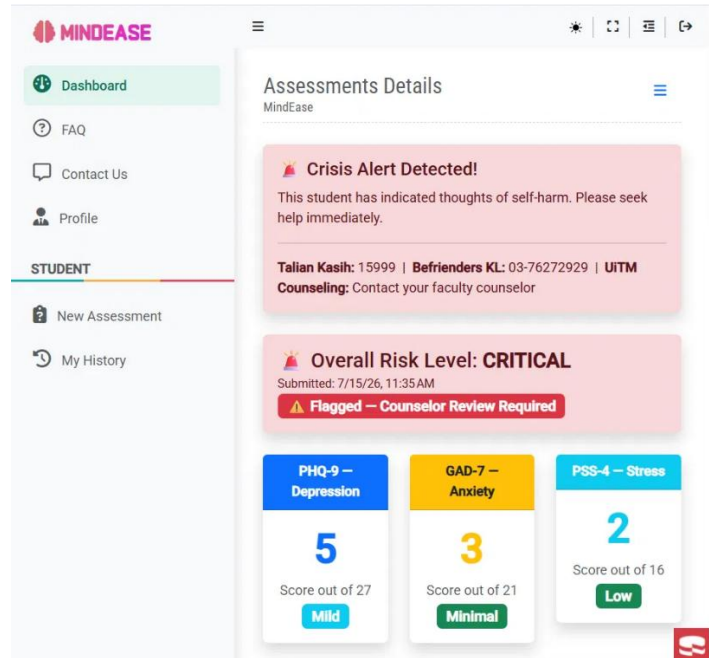


Figure 6.2.3: Assessment Details showing Crisis Alert banner, CRITICAL risk level, Flagged badge, and PHQ-9/GAD-7/PSS-4 score cards

6.2.4 My History (For Student)

Students can revisit all past submissions through the **Assessments List** page (My History), which displays a paginated table of every assessment the student has submitted, allowing them to track how their wellbeing changes over time.

Columns displayed:

- **#** — row number
- **Student** — student name and email
- **PHQ-9** — depression score
- **GAD-7** — anxiety score
- **PSS-4** — stress score
- **Overall Risk** — color-coded badge (CRITICAL, HIGH, MILD, LOW)
- **Status** — shows OK, Flagged, or Crisis
- **Submitted** — date and time of submission
- **Actions** — View button (eye icon) only

Student Data-Privacy Restrictions:

- My History is filtered by the logged-in user's ID — a student sees only their own records, never other students' assessments
- Edit and Delete actions are hidden from students — submitted assessments are read-only
- The Search & Filter panel is hidden from students, as it is only meaningful for counselors triaging many records
- The + Add Note button is hidden from students, since clinical notes may only be created by counselors
- The **+ New Assessment** button is visible, allowing students to submit a new check-in directly from this page

How to Use:

1. Log in as a student and click **My History** in the sidebar
2. Your past assessments are listed in a table showing PHQ-9, GAD-7, PSS-4 scores, overall risk badge, and status
3. Click the **View** button on any row to open the full Assessment Details page, including score cards and any counselor feedback
4. Use the **Prev / Next** pagination at the bottom to browse older submissions

#	Student	PHQ-9	GAD-7	PSS-4	Overall Risk	Status	Submitted	Actions
1	Ahmad Rayyan Iman student@mendease.com	13	11	10	CRITICAL	Crisis	6/21/26, 10:24PM	View
2	Ahmad Rayyan Iman student@mendease.com	2	2	2	LOW	✓ OK	6/22/26, 7:20AM	View
3	Ahmad Rayyan Iman student@mendease.com	16	12	12	HIGH	Flagged	6/22/26, 11:15AM	View
4	Ahmad Rayyan Iman student@mendease.com	9	7	3	MILD	✓ OK	7/11/26, 3:01PM	View
5	Ahmad Rayyan Iman student@mendease.com	9	4	6	CRITICAL	Crisis	7/11/26, 10:37PM	View
6	Ahmad Rayyan Iman student@mendease.com	12	4	4	CRITICAL	Crisis	7/13/26, 1:09PM	View
7	Ahmad Rayyan Iman student@mendease.com	0	0	0	LOW	✓ OK	7/14/26, 11:23AM	View
8	Ahmad Rayyan Iman student@mendease.com	5	3	2	CRITICAL	Crisis	7/15/26, 11:35AM	View

Page 1 of 1, showing 8 of 8 total

Figure 6.2.4: My History (Assessments List) student's own submissions

6.2.5 All Assessments (For Counselor & Admin)

Counselors and admins access all student submissions through the **Assessments List** page, which is divided into three tabs:

List Tab — main view showing all submissions in a table with the following columns:

- **#** — row number
- **Student** — student name and email
- **PHQ-9, GAD-7, PSS-4** — computed scores
- **Overall Risk** — color-coded badge (CRITICAL, HIGH, MILD, LOW)
- **Status** — ✓ OK, Flagged, or Crisis
- **Submitted** — date and time
- **Review** — Pending or Reviewed
- **Actions** — three buttons: View, Add Note (green), Delete (red)

Export Tab — provides a **PDF Download** button to export the full assessment list as a PDF report.

Search & Filter Panel (right side, visible on all tabs):

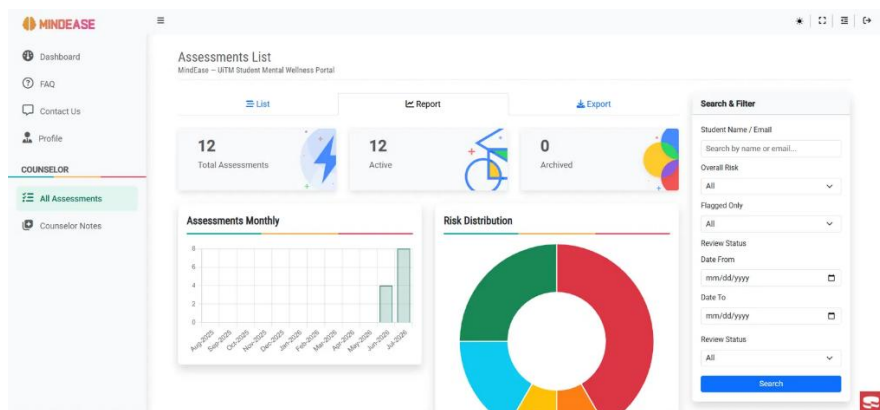
- Search by Student Name / Email
- Filter by Overall Risk (All / CRITICAL / HIGH / MODERATE / MILD / LOW)
- Filter by Flagged Only
- Filter by Review Status
- Date From and Date To range filter
- **Search** button to apply filters

How to use:

1. Log in as a counselor and click **All Assessments** in the sidebar
2. On the **List** tab, use the Search & Filter panel on the right to locate priority cases — filter by CRITICAL or Flagged Only to triage urgently
3. Click the **View** button to open the full Assessment Details page with student information, all 25 responses, computed scores, and previous notes
4. Click the **Add Note** button to add a counselor note directly from the list
5. Switch to the **Report** tab to view submission trends and risk distribution charts
6. Switch to the **Export** tab and click **PDF Download** to export the assessments list

#	Student	PHQ-9	GAD-7	PSS-4	Overall Risk	Status	Submitted	Review	Actions
1	Administrator admin@mindease.com	10	7	8	CRITICAL	Crisis	6/26/26, 8:08 PM	Pending	[Icons]
2	Ahmad Rayhan Iman student@mindease.com	13	11	10	CRITICAL	Crisis	6/21/26, 10:24 PM	Pending	[Icons]
3	Ahmad Rayhan Iman student@mindease.com	2	2	2	LOW	OK	6/22/26, 7:23 AM	Pending	[Icons]
4	Ahmad Rayhan Iman student@mindease.com	16	12	12	HIGH	Flagged	6/22/26, 11:15 AM	Reviewed	[Icons]
5	Ahmad Rayhan Iman student@mindease.com	9	7	3	MED	OK	7/11/26, 3:21 PM	Pending	[Icons]
6	Ahmad Rayhan Iman student@mindease.com	9	4	6	CRITICAL	Crisis	7/11/26, 10:57 PM	Pending	[Icons]
7	Ahmad Rayhan Iman student@mindease.com	12	4	4	CRITICAL	Crisis	7/13/26, 1:35 PM	Reviewed	[Icons]
8	Ahmad Rayhan Iman student@mindease.com	0	0	0	LOW	OK	7/14/26, 11:23 AM	Pending	[Icons]
9	Dania Fakhira	4	6	1	MED	OK	7/15/26, 8:10 AM	Reviewed	[Icons]

Figure 6.2.5.1: All Assessments — List tab



6.3 COUNSELOR NOTES MODULE

The CounselorNotes module allows counselors to document their clinical follow-up on each assessment. Notes are linked to a specific assessment, forming a traceable chain: Student submits assessment → Counselor reviews in All Assessments → Counselor adds a note → note appears inside that assessment's detail view.

The module is divided into three tabs — **List**, **Report**, and **Export** — with a persistent **Search & Filter** panel and **Action Legend** on the right side.

List Tab — columns displayed:

- **#** — row number
- **Student / Assessment** — student name with the assessment's risk badge
- **Counselor** — counselor ID who wrote the note
- **Action Taken** — color-coded badge showing the action type
- **Follow Up** — scheduled follow-up date
- **Status** — Active or Closed
- **Created** — date and time the note was created
- **Actions** — three buttons: View, Edit, Delete

Action Taken types (with color badges):

Action	Badge Color
No Action	Grey
Follow up	Blue
Referred	Orange
Contacted	Teal
Crisis Intervention	Red

Report Tab:

- Three stat cards: Total Notes, Active, Archived
- Notes Monthly bar chart
- Notes by Status donut chart (Active / Closed / Archived)

Export Tab:

- PDF Download button to export the full counselor notes list

Export Tab:

- PDF Download button to export the full counselor notes list

Note Form Fields (Add / Edit):

- **Select Assessment**— dropdown showing Student Name — Date [Risk Level] format
- **Counselor**— auto-filled with the logged-in counselor's name (read-only)
- **Clinical Note**— free-text area for observations and clinical remarks
- **Action Taken**— select from No Action / Follow Up / Referred / Contacted / Crisis Intervention
- **Follow Up Date**— date picker

Note Detail View (CounselorNotes Details):

- NOTE INFORMATION panel: Note ID, Assessment link with risk badge, Counselor ID, Action Taken, Follow Up Date, Status, Created timestamp
- CLINICAL NOTE panel: full text of the clinical note
- QUICK ACTIONS panel: Edit This Note, View Assessment, Add Another Note, Back to List

Key Features:

- Full CRUD for notes (create, view, edit, delete)
- Counselor field is auto-filled from the logged-in user — cannot be changed, ensuring accountability
- Students can read counselor feedback on their own submissions but cannot create, edit, or delete notes
- Note status (Active / Closed) is independent from Action Taken type

How to Use (Adding a Note — Counselor):

1. Click **Counselor Notes** in the sidebar, then click **+ New Note**, or click the Add Note button directly from the All Assessments list
2. Select the related assessment from the dropdown — it shows Student Name — Date [Risk Level] so you can identify the case

3. Write your clinical observations in the **Clinical Note** text area
4. Select the **Action Taken** (Follow Up, Referred, Contacted, Crisis Intervention, or No Action)
5. Set a **Follow Up Date** if a scheduled follow-up is needed
6. Click **Submit** — the note is saved and the student can read the feedback in their Assessment Details page
7. To edit a note, return to the CounselorNotes List and click the Edit button on the relevant row

#	STUDENT / ASSESSMENT	COUNSELOR	ACTION TAKEN	FOLLOW UP	STATUS	CREATED	ACTIONS
1	Ahmad Raiyan Iman	10:2	Crisis Intervention	22 Jun 2026	Active	6/22/26, 3:18 AM	[Edit] [Delete]
2	Ahmad Raiyan Iman	10:2	Follow Up	22 Jun 2026	Active	6/22/26, 3:44 AM	[Edit] [Delete]
3	Ahmad Raiyan Iman	10:2	Follow Up	17 Jul 2026	Active	7/14/26, 12:15 AM	[Edit] [Delete]
4	Dania Fakhira	10:2	Contacted	21 Jul 2026	Active	7/15/26, 12:35 AM	[Edit] [Delete]
5	Dania Fakhira	10:2	Referred	23 Jul 2026	Active	7/15/26, 1:25 AM	[Edit] [Delete]
6	Dania Fakhira	10:2	No Action	—	Closed	7/15/26, 3:52 AM	[Edit] [Delete]
7	Ahmad Raiyan Iman	10:2	Contacted	16 Jul 2026	Active	7/15/26, 3:58 AM	[Edit] [Delete]

Figure 6.3.1: CounselorNotes List tab

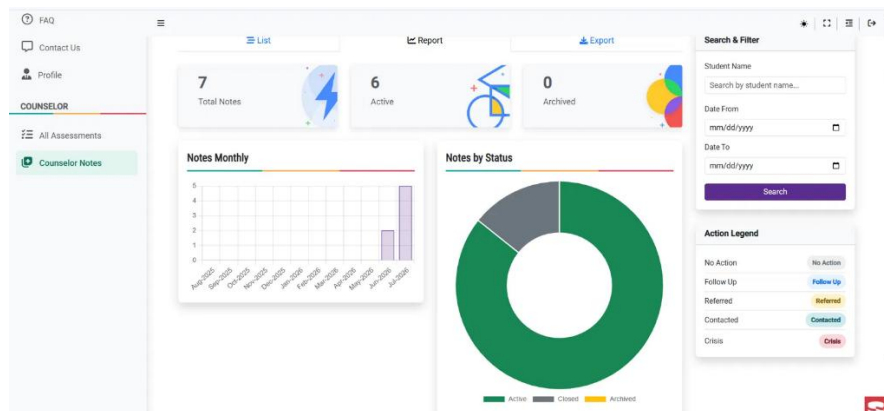


Figure 6.3.2: CounselorNotes Report tab

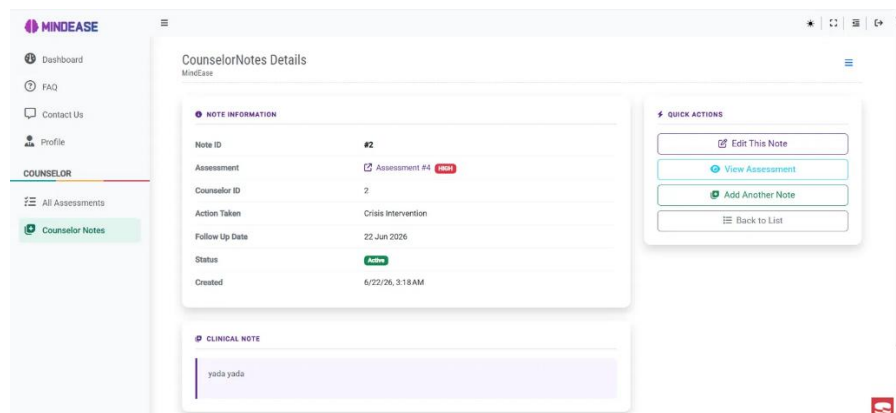


Figure 6.3.3: CounselorNotes Details page

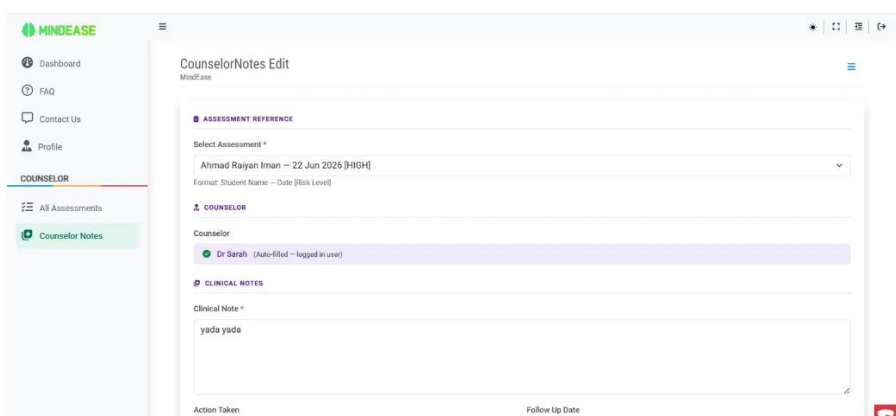


Figure 6.3.4: CounselorNotes Edit

6.4 QUESTIONS MANAGEMENT MODULE (ADMIN ONLY)

Administrators manage the 25-question assessment bank through the **Questions List** page. The module follows the same List / Report / Export tab structure as other modules, with a **Section Legend** on the right panel colour-coding each section.

List Tab — columns displayed:

Column	Description
Code	Question code (A1–A9, B1–B7, C1–C4, D1–D5)
Section	Colour-coded badge: A–Depression / B–Anxiety / C–Stress / D–Wellbeing
Question	Full question text
Max score	Maximum scoreable value per question
Crisis?	Red "Yes" badge only for A9; dash for all others

Order	Display sequence in the assessment form
Status	Active / Inactive
Action	View, Edit, Delete

Admin Capabilities:

- **Create** — add new questions with section, code, text, response type, max score, is_crisis_trigger, order, and status
- **Read** — view all 25 questions with full field details and related responses
- **Update** — reword or reconfigure a question without breaking historical response records
- **Delete** — remove a question with confirmation

How to Use:

1. Log in as admin and click **Questions** in the sidebar
2. Browse all 25 questions across 3 pages (10 per page)
3. Click **Edit** to modify a question's wording, max score, or order
4. Click **View** to see the question details and all recorded responses for that question

MINDEASE

Questions List

MINDEASE

List Report Export

#	CODE	SECTION	QUESTION	MAX SCORE	CRISIS?	ORDER	STATUS	ACTIONS
1	A1	A - Depression	Little interest or pleasure in doing things	3	—	1	Active	[View] [Edit] [Delete]
2	A2	A - Depression	Feeling down, depressed, or hopeless	3	—	2	Active	[View] [Edit] [Delete]
3	A3	A - Depression	Trouble falling or staying asleep, or sleeping too much	3	—	3	Active	[View] [Edit] [Delete]
4	A4	A - Depression	Feeling tired or having little energy	3	—	4	Active	[View] [Edit] [Delete]
5	A5	A - Depression	Poor appetite or overeating	3	—	5	Active	[View] [Edit] [Delete]
6	A6	A - Depression	Feeling bad about yourself or that you are a failure	3	—	6	Active	[View] [Edit] [Delete]
7	A7	A - Depression	Trouble concentrating on things such as studying	3	—	7	Active	[View] [Edit] [Delete]
8	A8	A - Depression	Moving or speaking slowly, or being fidgety and restless	3	—	8	Active	[View] [Edit] [Delete]
9	A9	A - Depression	Thoughts of hurting yourself or being better off dead	3	Yes	9	Active	[View] [Edit] [Delete]
10	B1	B - Anxiety	Feeling nervous, anxious, or on edge	3	—	10	Active	[View] [Edit] [Delete]

Page 1 of 3, showing 10 of 25 total

Section Legend

PHQ-9 A - Depression

GAD-7 B - Anxiety

PSS-4 C - Stress

General D - Wellbeing

Figure 6.4.1: Questions List

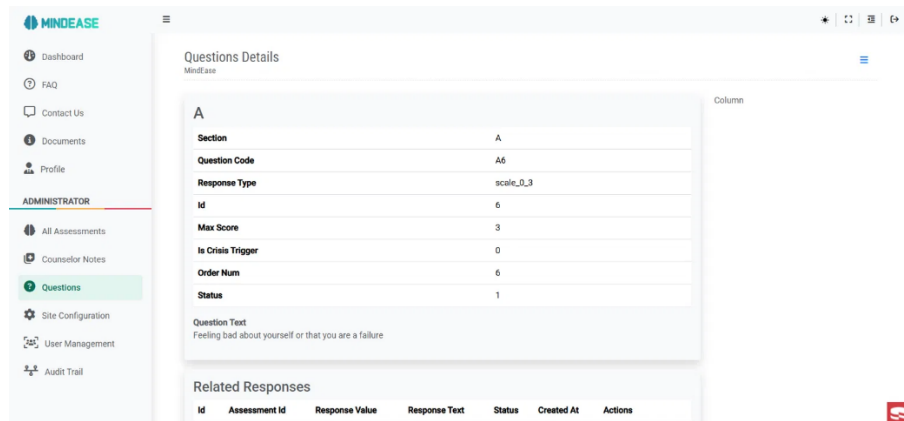


Figure 6.4.2: Questions Details page

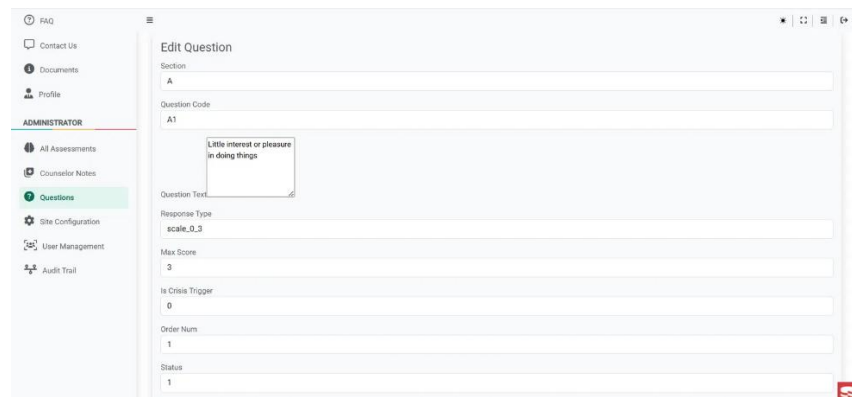


Figure 6.4.3: Edit Question

6.5 DASHBOARD MODULE (ROLE-AWARE)

The Dashboard is the landing page after login and renders different content depending on the user's role. The dashboard controller checks the identity's role and loads the appropriate view and data queries.

Counselor / Admin Dashboard:

Three statistics cards at the top:

- **Total Assessments** (blue)
- **Flagged Cases** (red)
- **Critical Risk** (orange)

Three Chart.js charts below:

- **Risk Distribution** donut chart — breaks down all submissions by overall risk level with a legend: Critical (red), High (orange), Moderate (yellow), Mild (teal), Low (green)
- **Assessments (Last 6 Months)** bar chart — shows monthly submission volume in purple bars for the past 6 months
- **Average Scores** horizontal bar chart — compares system-wide average PHQ-9 (blue), GAD-7 (orange), and PSS-4 (teal) scores on a scale of 0–10

Student Dashboard:

- **Purple gradient welcome banner** — personalised greeting example, "Hi, Ahmad Raiyan Iman" with the subtext "How are you feeling today? Take a moment to check in on your mental wellbeing." and a **New Assessment** button on the right
- Three statistics cards: **My Total Assessments** (purple), **Flagged Results** (red), **Latest Risk Level** shown as a full color-coded badge (e.g. CRITICAL in red)
- **My Score Trend** line chart — plots the student's PHQ-9 (blue), GAD-7 (orange), and PSS-4 (teal) scores across all their submissions so they can visually track changes over time
- **Recent Assessments** panel — lists the 5 most recent submissions with date, time, risk badge, and a View button

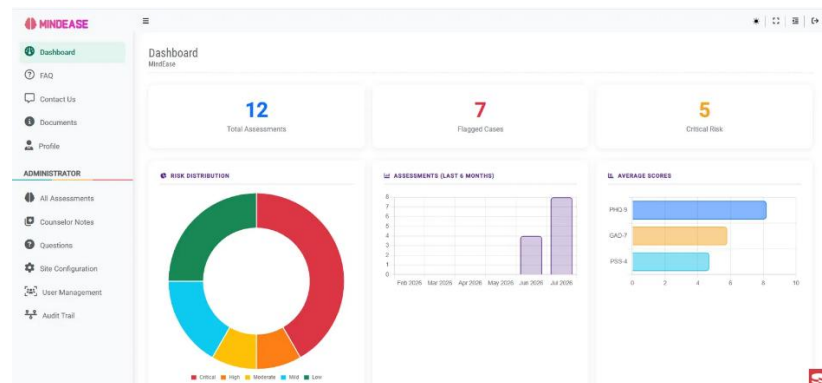


Figure 6.5.1: Admin dashboard

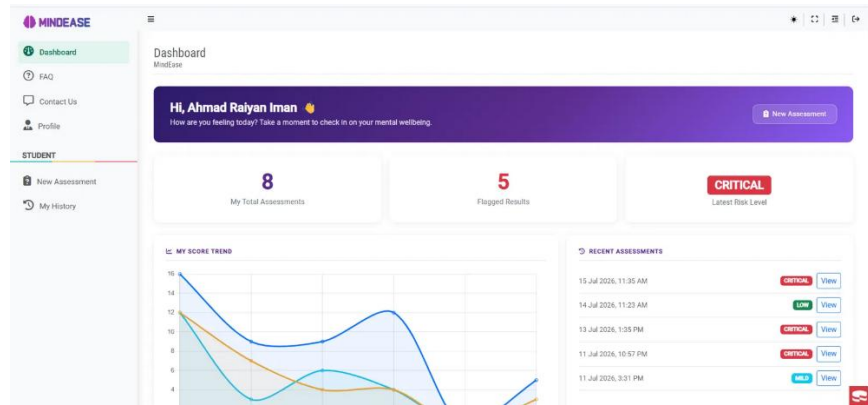


Figure 6.5.2: Counselor dashboard

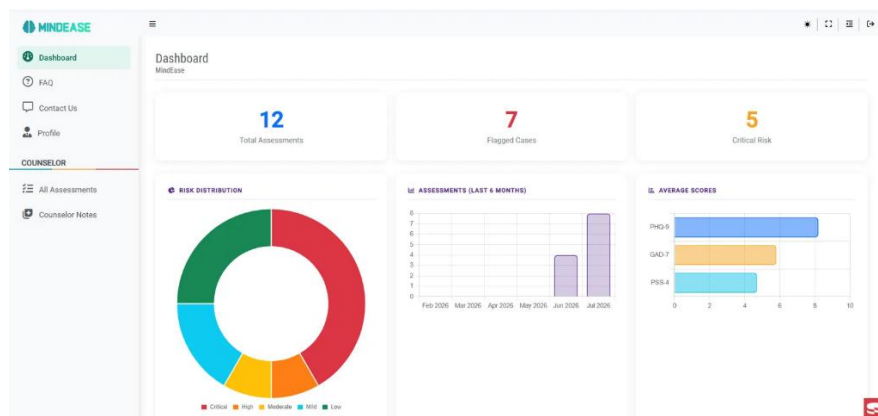


Figure 6.5.3: Student dashboard

6.6 PDF REPORT GENERATION

MindEase uses the CakePDF plugin (DomPDF engine) to generate two distinct PDF reports, accessible via the Export tab in their respective modules.

Report 1 — MindEase Student Wellness Report

Generated from the All Assessments Export tab. Contents:

- Title: *MindEase — Student Wellness Report*, Universiti Teknologi MARA (UiTM)
- Generated date/time and total record count
- Four summary stat cards: Total Assessments, Flagged Cases, Critical Risk, High Risk
- Full assessment table with columns: #, User ID, PHQ-9 score + Depression level, GAD-7 score + Anxiety level, PSS-4 score + Stress level, Overall Risk badge, Flagged (YES/NO), Submitted date/time

- Footer: "MindEase — UiTM Student Mental Wellness Assessment Portal | **CONFIDENTIAL**. This report is generated automatically by the system. For clinical use only."

Report 2 — MindEase Assessment Questions Report

Generated from the Questions module Export tab. Contents:

- Title: *MindEase — Assessment Questions Report*, UiTM Student Mental Wellness Portal
- Generated date/time
- Section summary row: Section A (PHQ-9), Section B (GAD-7), Section C (PSS-4), Section D (Wellbeing), Total question count
- Full questions table with columns: #, Code, Section badge, Question text, Response Type, Max Score, Crisis? (Yes badge for A9 only), Status

How to Use:

1. Log in as admin or counselor
2. For the **Wellness Report** — go to All Assessments → Export tab → click **PDF Download**
3. For the **Questions Report** — go to Questions → Export tab → click **PDF Download**
4. The PDF opens in the browser and can be saved or printed

MindEase — Student Wellness Report

Universiti Teknologi MARA (UiTM)

Generated: 15 July 2026, 06:36 PM | Total Records: 12

12

Total Assessments

7

Flagged Cases

5

Critical Risk

1

High Risk

#	User ID	PHQ-9	Depression	GAD-7	Anxiety	PSS-4	Stress	Overall Risk	Flagged	Submitted
1	1	10	Moderate	7	Mild	8	Moderate	CRITICAL	YES	20/06/2026 08:08 PM
2	3	13	Moderate	11	Moderate	10	High	CRITICAL	YES	21/06/2026 10:24 PM
3	3	2	Minimal	2	Minimal	2	Low	LOW	No	22/06/2026 07:20 AM
4	3	16	Moderately Severe	12	Moderate	12	High	HIGH	YES	22/06/2026 11:15 AM
5	3	9	Mild	7	Mild	3	Low	MILD	No	11/07/2026 03:31 PM
6	3	9	Mild	4	Minimal	6	Moderate	CRITICAL	YES	11/07/2026 10:57 PM
7	3	12	Moderate	4	Minimal	4	Low	CRITICAL	YES	13/07/2026 01:35 PM
8	3	0	Minimal	0	Minimal	0	Low	LOW	No	14/07/2026 11:23 AM
9	6	4	Minimal	9	Mild	1	Low	MILD	No	15/07/2026 08:30 AM
10	3	5	Mild	3	Minimal	2	Low	CRITICAL	YES	15/07/2026 11:35 AM
11	6	4	Minimal	3	Minimal	2	Low	LOW	No	15/07/2026 11:42 AM
12	6	14	Moderate	8	Mild	6	Moderate	MODERATE	YES	15/07/2026 11:43 AM

MindEase — UiTM Student Mental Wellness Assessment Portal | CONFIDENTIAL

This report is generated automatically by the system. For clinical use only.

Figure 6.6.1: MindEase Student Wellness Report PDF

MindEase — Assessment Questions Report							
UiTM Student Mental Wellness Portal Generated on 16 Jul 2026, 03:32 PM							
SECTION A (PHQ-9)		SECTION B (GAD-7)		SECTION C (PSS-4)		SECTION D (WELLBEING)	TOTAL
9		7		4		0	20
#	CODE	SECTION	QUESTION	RESPONSE TYPE	MAX SCORE	CRISIS?	STATUS
1	A1	A — Depression	Little interest or pleasure in doing things	scale_0_3	3	—	Active
2	A2	A — Depression	Feeling down, depressed, or hopeless	scale_0_3	3	—	Active
3	A3	A — Depression	Trouble falling or staying asleep, or sleeping too much	scale_0_3	3	—	Active
4	A4	A — Depression	Feeling tired or having little energy	scale_0_3	3	—	Active
5	A5	A — Depression	Poor appetite or overeating	scale_0_3	3	—	Active
6	A6	A — Depression	Feeling bad about yourself or that you are a failure	scale_0_3	3	—	Active
7	A7	A — Depression	Trouble concentrating on things such as studying	scale_0_3	3	—	Active
8	A8	A — Depression	Moving or speaking slowly, or being fidgety and restless	scale_0_3	3	—	Active
9	A9	A — Depression	Thoughts of hurting yourself or being better off dead	scale_0_3	3	Yes	Active
10	B1	B — Anxiety	Feeling nervous, anxious, or on edge	scale_0_3	3	—	Active

Figure 6.6.2: MindEase Assessment Questions Report PDF

6.7 USERS MANAGEMENT AND SITE CONFIGURATION (ADMIN ONLY)

Administrators manage all accounts through the Users module, including creating counselor accounts and assigning roles (admin, counselor, student). The Site Configuration panel, provided by reCRUD, was used to rebrand the entire system with the MindEase identity: system name, abbreviation, slogan, organization name, domain, notification email, and author.

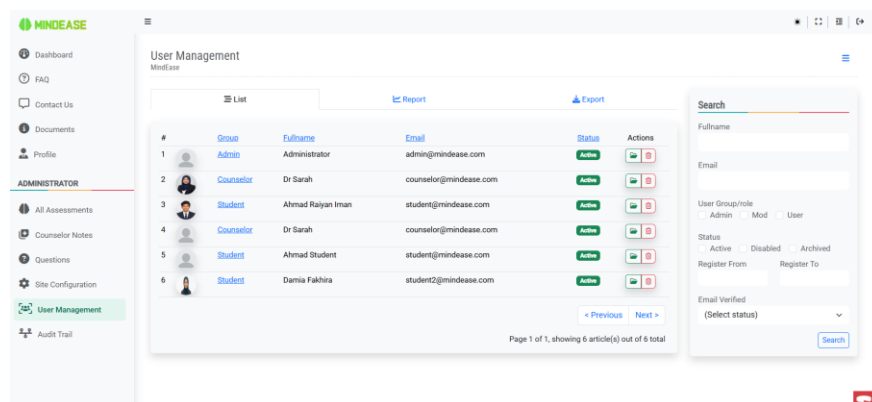


Figure 6.7: Users Management page with role assignment

6.8 INFORMATION PAGES

6.8.1 FAQ Page

The FAQ page displays real MindEase content seeded into the faqs table, organised into a two-column grid of category cards. Each question is displayed as a collapsible accordion item like clicking the + icon expands the answer, and clicking again collapses it.

FAQ Categories and Questions:

Category	Questions
General	What is MindEase? / Is my information kept confidential? / Who can use MindEase?
Assessment	What is PHQ-9? / What is GAD-7? / What is PSS-4? / What does a CRITICAL or HIGH score mean? / How often should I complete an assessment?
Account	How do I change my password? / How do I update my profile information? / I forgot my password. What should I do?
Support	Who can I contact in an emergency? / How will a counselor contact me?

How to Use:

1. Click FAQ in the sidebar or the FAQ quick link on the login page
2. Browse the four category cards
3. Click any question's + button to expand its answer; click again to collapse

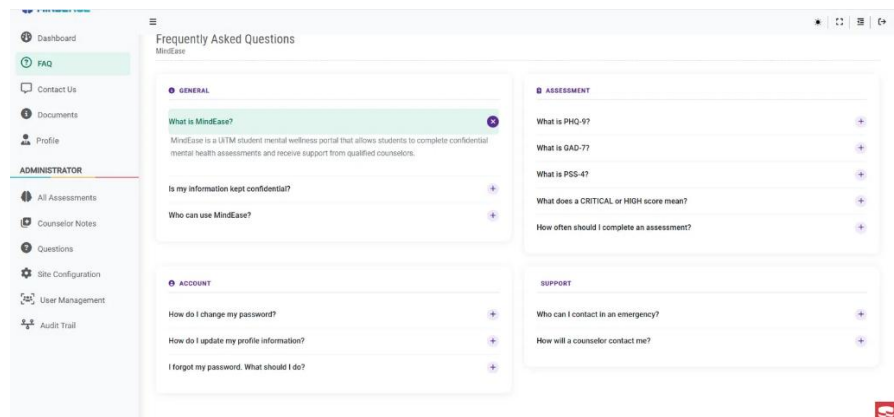


Figure 6.8.1: FAQ page

6.8.2 Contact Us Page

The Contact Us page provides two things on one screen — a message form on the left and the real **Pusat Psikologi & Kaunseling UiTM** contact details on the right, along with an embedded map.

Send Us a Message form fields:

- **Ticket ID** — auto-generated unique reference code example “Xez-ytPq” shown at the top so the student can save it to check for a response later via the **Check Response** button
- **Subject** — brief topic of the message
- **Your Name** and **Email Address**
- **Message** — free-text area
- **Reset** and **Send Message** buttons

Right panel — Pusat Psikologi & Kaunseling UiTM:

- Address: Aras 5, Kompleks Budisiswa, Jabatan Hal Ehwal Pelajar, UiTM, 40450 Shah Alam, Selangor
- Phone: 03-5544 2630
- Email: ppk@uitm.edu.my
- Hours: Monday–Friday, 8:00 AM–5:00 PM
- Websites: counselling.uitm.edu.my / counselling2u.uitm.edu.my
- **Emergency:** Talian Kasih 15999 / Befrienders KL 03-7627 2929
- Social media links and embedded Google Map with Open in Maps link

How to Use:

1. Click **Contact Us** in the sidebar or the quick link on the login page
2. Note the auto-generated **Ticket ID** — save it to follow up later
3. Fill in Subject, Name, Email Address, and Message
4. Click **Send Message** — your message is stored for follow-up

5. To check if a response has been given, return to Contact Us and click **Check Response**, entering your Ticket ID

MINDEASE

Dashboard
FAQ
Contact Us
Documents
Profile
ADMINISTRATOR
All Assessments
Counselor Notes
Questions
Site Configuration
User Management
Audit Trail

Contact
MindEase

SEND US A MESSAGE

TICKET ID
Xez-ytPq
Save this to check your response later

Check Response

Subject
What is your message about?

Your Name
Full name

Email Address
your@email.com

Message
Write your message here...

Send Message

Pusat Psikologi & Konseling UTM
Area 5, Kompleks Budidarma,
Jabatan Hal Ehwal Pelajar,
UTM, 40450 Shah Alam, Selangor
03-5544 2630
pkip@utm.edu.my
Monday - Friday | 9:00 AM - 5:00 PM
counseling.utm.edu.my
counseling2u.utm.edu.my
Emergency?
Talian Kasih: 19999
Berfrenders KL: 03-7427 2529

FOLLOW US

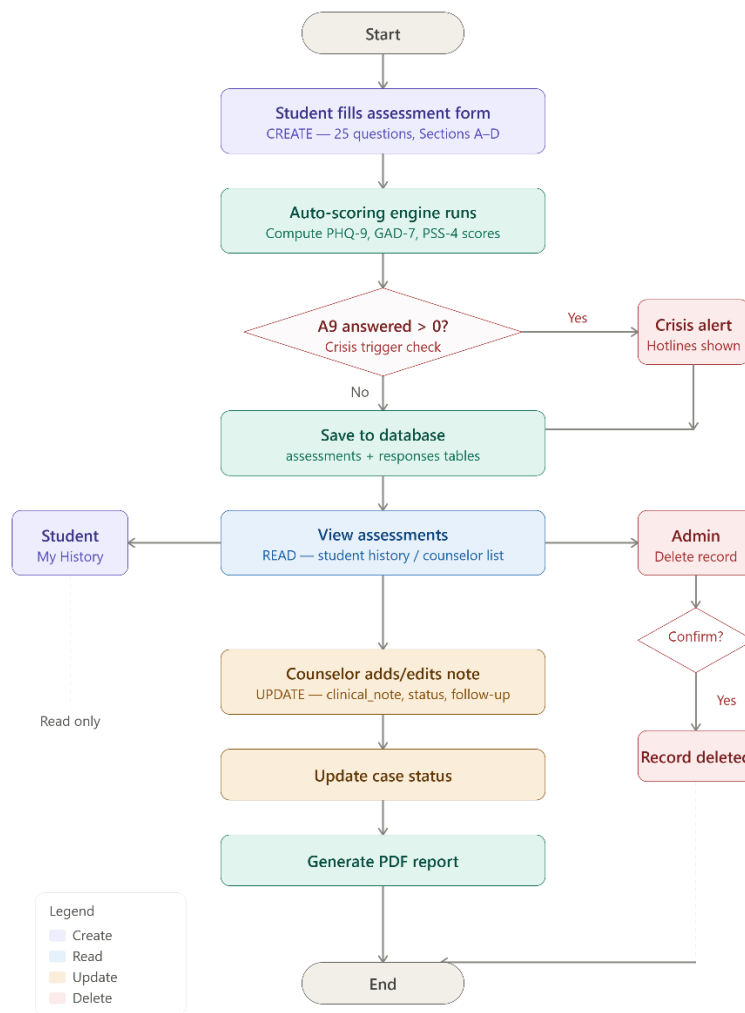
Open in Maps

Shah Alam

Figure 6.8.2: Contact Us page

7.0 WORKFLOW OF FORM (ILLUSTRATION OF THE CRUD CYCLE)

7.1 CRUD WORKFLOW DIAGRAM



7.2 DETAILED CRUD PROCESS

CREATE — Student fills the 25-question assessment form and submits. The AssessmentsController runs the auto-scoring engine, computes PHQ-9, GAD-7, and PSS-4 scores, checks question A9 for the crisis trigger, and saves one record to the assessments table plus 25 records to the responses table.

READ — Both students and counselors view assessment records. Students access only their own records via My History (filtered by user_id). Counselors and admins access all records via the All Assessments list with search and filter options.

UPDATE — Counselors add or edit a CounselorNote linked to an assessment, documenting the clinical_note, action_taken, follow_up_date, and status. This updates the case without modifying the original assessment record.

DELETE — Admins can delete invalid or test records from the All Assessments list after a confirmation prompt. Related responses are removed together with the assessment.

7.3 CRUD OPERATIONS FOR OTHER MODULES

CounselorNotes Module (Counselor)

Operation	Action	Result
CREATE	Counselor submits a new note linked to an assessment	Note saved with clinical_note, action_taken, follow_up_date, and status
READ	Counselor views the CounselorNotes List; student views notes on their own Assessment Details page	Notes displayed with action badge, status, and follow-up date
UPDATE	Counselor edits an existing note via the Edit form	clinical_note, action_taken, follow_up_date, and status updated
DELETE	Counselor deletes a note with confirmation	Note removed from the system

Questions Module (Admin)

Operation	Action	Result
CREATE	Admin adds a new question to the question bank	Question available in the assessment form, ordered by order_num
READ	Admin views the Questions List across 3 pages; system loads questions dynamically to render the form	All 25 questions displayed with section badge, max score, and crisis flag

UPDATE	Admin edits question wording, max_score, order_num, or status	Updated question reflects immediately in the assessment form
DELETE	Admin deletes a question with confirmation	Question removed; historical responses remain in the responses table

Users Module (Admin)

Operation	Action	Result
CREATE	Admin creates a new counselor or student account with role assignment	New user can log in with the assigned role
READ	Admin views the User Management list with all accounts and roles	All accounts displayed with status and group
UPDATE	Admin edits user details, role, or status	Account updated; extended fields (student_id, faculty, program, year_of_study) editable for student accounts
DELETE	Admin removes an account with confirmation	User account deleted from the system

8.0 TEAM ROLES AND CONTRIBUTIONS

No	Name	Matric No.	Role	Contribution
1	PUTERI NURYASMIN FARHANA BINTI MEGAT MOHD ZULKARNAIN	2024402302	Project Manager	Project coordination, system scope and database schema design (ERD, ALTER TABLE, entity relationships), assessment auto-scoring logic in AssessmentsController, crisis trigger implementation, role-based access control, quality assurance and testing across all three roles
2	AHMAD RAIYAN IMAN BIN AZWAN	2024413898	Backend Developer	Assessments module full CRUD, responses saving logic, counselor notes module, PDF report generation using CakePDF/DomPDF (Wellness Report and Questions Report)
	NUR AZWA BINTI ABDUL AZIZ	2024665854	Backend Developer	Questions (question bank) module, dashboard data queries (Chart.js integration), FAQ content seeding, Contact Us module with ticket ID system
	ASMA AINA BINTI JURAIME	2024804792	Frontend Developer & Documentation	Clinical assessment form template with 4-step progress indicator, custom split-panel login page with Vanta.js, profile page redesign with QR code, role-aware sidebar and dashboard, MindEase colour theme, report writing, ERD documentation, screenshots and visual assets

9.0 CONTACT INFORMATION

9.1 SYSTEM CONTACT

Contact Type	Details
System	MindEase UiTM Student Mental Wellness Portal
Organization	Universiti Teknologi MARA
System Email	mindease@uitm.edu.my
Version	MindEasev1.1

9.2 CRISIS SUPPORT RESOURCES

As a mental wellness system, MindEase prominently displays the following support lines to students on the assessment result page and Contact Us page:

Resource	Contact
Talian Kasih	15999
Befrienders Kuala Lumpur	03-7627 2929
Pusat Psikologi & Kaunseling UiTM	03-5544 2630
PPK Email	ppk@uitm.edu.my
PPK Website	Counselling.uitm.edu.my
PPK Address	Aras 5, Kompleks Budisiswa, Jabatan Hal Ehwal Pelajar, UiTM, 40450 Shah Alam, Selangor
Office Hours	Monday-Friday, 8:00 am – 5:00 pm

9.3 ACADEMIC CONTACT

- Course Lecturer: Dr. Muhammad Asyraf bin Wahi Anuar
- Email: muham6853@uitm.edu.my

10.0 CONCLUSION AND REFLECTION

10.1 PROJECT SUMMARY

The MindEase system was successfully developed to digitalise the student mental wellness check-in process at UiTM. The system delivers three main outcomes:

1. **Structured Self-Assessment** — Students complete a clinically grounded 25-question assessment (PHQ-9, GAD-7, PSS-4, and General Wellbeing) online instead of on paper, and instantly receive their scores and risk level.
2. **Automated Triage** — The scoring and risk engine automatically flags moderate, high, and critical cases, with an immediate crisis alert pathway for any indication of self-harm ideation, ensuring the most urgent students surface first.
3. **Counselor Case Management** — Counselors review submissions in a color-coded dashboard, document clinical notes with follow-up dates, and track case status from Pending through to Resolved or Referred, while admins oversee users, the question bank, and PDF reporting.

Role-based access control keeps sensitive assessment data restricted: students see only their own records, counselors see clinical data, and only administrators manage users and system configuration.

10.2 TECHNICAL ACHIEVEMENTS

Throughout this project, our team successfully implemented:

- MVC Architecture using the reCRUD framework (CakePHP 4) for clean code organisation
- A normalised five-table database with proper foreign key relationships (users, questions, assessments, responses, counselor_notes)
- A question-bank design in which the form is rendered dynamically from the database, separating questions (the exam paper) from responses (the answer script)
- An automatic clinical scoring and risk engine implemented in the AssessmentsController, including the A9 crisis trigger

- Role-based access control differentiating student, counselor, and admin capabilities in the sidebar, dashboard, and per-action buttons (Edit/Delete/Add Note hidden from students), with URL-level redirects for restricted pages
- Data privacy by design — My History filters records by the logged-in user so students only ever see their own assessments
- PDF generation using the CakePDF plugin with the DomPDF engine for two distinct reports: the Student Wellness Report and the Assessment Questions Report
- Dashboard data visualisations with Chart.js — Risk Distribution donut chart, Assessments Last 6 Months bar chart, Average Scores horizontal bar chart (counselor/admin), and My Score Trend multi-line chart tracking PHQ-9, GAD-7, and PSS-4 across submissions (student)
- A fully rebranded user experience: custom split-panel login page with Vanta.js animated background, purple theme, redesigned profile page with QR code, and a categorised accordion FAQ with real mental wellness content
- Full CRUD operations for all major data entities

10.3 CHALLENGES FACED AND SOLUTIONS

Challenge	Solution
Column naming mismatch — reCRUD's auditing and templates expected a column named created, but our tables used created_at, causing errors on the index pages	Renamed the columns using ALTER TABLE ... CHANGE created_at created DATETIME so the framework conventions were satisfied
Baked controller variable errors — the code generator produced invalid variable names containing spaces (e.g. \$counselor note) in CounselorNotesController, resulting in Missing Controller / parse errors	Performed a careful find-and-replace to counselorNote in code contexts only, avoiding display strings, then retested every action
Role-based visibility — after customising the dashboard, all roles initially saw the counselor version	Made the dashboard and sidebar role-aware by checking the identity's role and rendering different content per role

Profile fields would not save — the new academic fields appeared in the form but were silently ignored on submit	Whitelisted student_id, faculty, program, and year_of_study in the User entity's _accessible array, as CakePHP's mass-assignment protection blocks fields that are not explicitly allowed
Permission error after profile update — students were redirected via referer() to a page they could not access	Changed the post-save redirect in UsersController to go directly to the user's own profile page
Missing PDF template — exporting counselor notes to PDF failed with a Missing Template error	Created the required pdf_list.php template under the module's pdf folder following the CakePDF convention
FAQ page showed placeholder content and wrong categories	Seeded real MindEase FAQ content into the faqs table via SQL and rebuilt the template as a categorised accordion (General, Assessment, Account, Support)
Virtual host issues — mindease.test did not resolve after setup	Restarted Laragon (Stop All / Start All) so it regenerated the virtual host, and used the localhost/mindease path as a fallback during development
Justifying the database design — explaining why scores are stored in assessments and why questions live in their own table	Documented that stored scores avoid recomputation on every dashboard load, and the question bank separates questions from answers just as an exam paper is separate from answer scripts

10.4 FUTURE IMPROVEMENTS

To further enhance the MindEase system, we recommend the following improvements for future development:

1. **Email Notifications** — alert counselors immediately when a critical case is submitted
2. **Assessment Reminders** — automated monthly prompts encouraging students to complete a new check-in to maintain their score trend data
3. **Counselor Assignment** — allow admins to assign specific students to specific counselors with caseload tracking

4. **Appointment Booking** — let flagged students book a counseling session directly from their result page
5. **48-hour Response Tracking** — measure counselor response time for high-risk cases
6. **Full Bilingual Interface** — extend the current system into a complete Bahasa Melayu version of the assessment and interface
7. **Deployment to a Live Server** — host the system on a real domain so features such as the profile QR code work from any device, not only on localhost
8. **Anonymised Analytics** — faculty-level wellness statistics for university management reporting
9. **Mobile Application** — a native app with push notification reminders for regular mental wellness check-ins

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